Name: ______ Cohort/Batch: ______ Date: ______ Score: ______

JDBC PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is JDBC and what are its core strengths? *Threat Model*

Q2 How does JDBC guarantee consistency and handle transactions? *Architecture*

Q3 What index types does JDBC support and when do you use each? *Configuration*

Q4 How do you design a schema or data model in JDBC? *Hardening*

Q5 How does JDBC scale horizontally? *SSO / IdP*

Q6 How do you monitor and tune slow queries in JDBC? *Observability*

Q7 What backup and restore strategies does JDBC support? *Lifecycle*

Q8 How do you handle schema migrations in JDBC? *Compliance*

Q9 What security features does JDBC provide? *Testing*

Q10 What are common anti-patterns when using JDBC in production? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 JDBC is a relational, document, key-value,

Mitigation

JDBC is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 JDBC provides a slow query log,

Observability

JDBC provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 JDBC provides either full ACID transactions

Primitives

JDBC provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 JDBC supports logical backups (export SQL

Automation

JDBC supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 JDBC offers B-tree, hash, full-text, spatial,

Hardening

JDBC offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in JDBC are managed

Governance

Schema migrations in JDBC are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in JDBC involves normalizing

Gotchas

Schema design in JDBC involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 JDBC supports role-based access control,

Assessment

JDBC supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 JDBC scales via replication (read replicas)

Federation

JDBC scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some JDBC implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.